



# EECS 841: Lecture 11 (case study: face detection)

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# Outline

## **Topic for Case Study:**

- Face detection

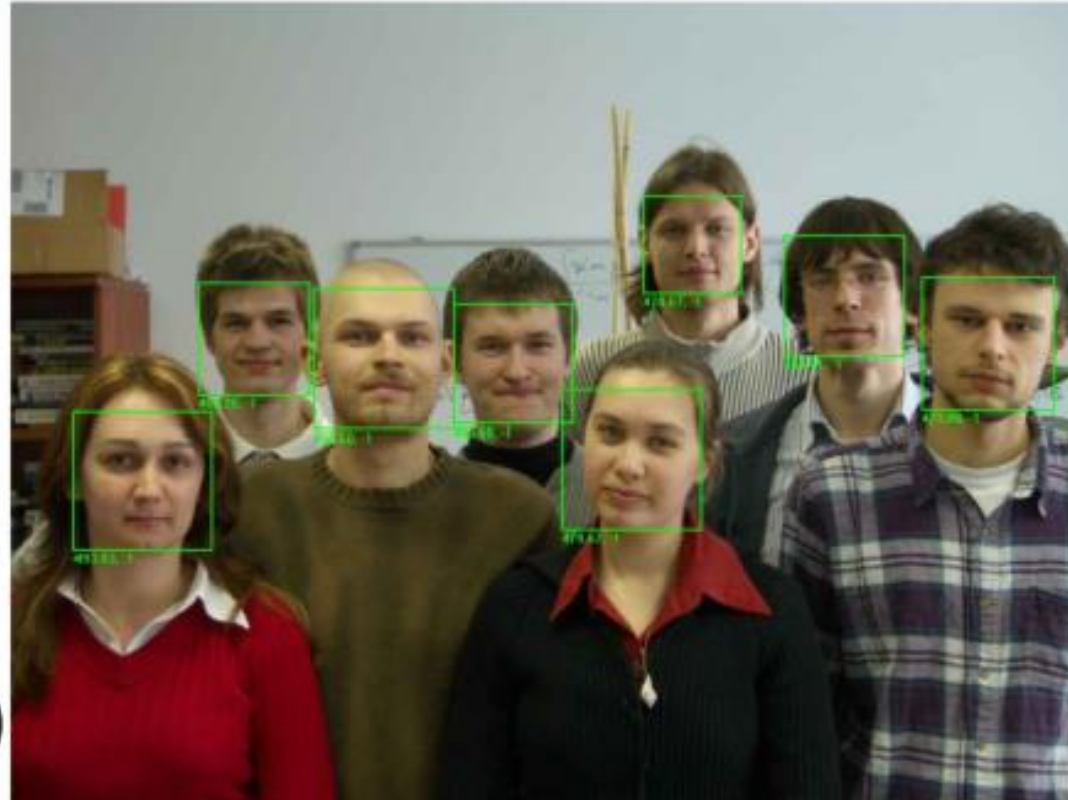
## Slide credit:

- Materials are used from “Introduction to Biometrics” text authors Arun Ross and Anil Jain as well as Dr. Walter Scheirer, Dr. Damon Woodard, and Ming-Hsuan Yang

# Face Detection: Introduction

**Definition:** Identify and locate human faces in an image regardless of their:

- Position
- Scale
- Orientation (in plane rotation)
- Pose (out-of plane rotation)
- Illumination



# Why is Face Detection Important?

## Security application:

- First step for any fully automatic face recognition system
- First step in many surveillance systems



Example Application: Facial Recognition

## Non-security application:



Example Application: Facial Motion Imprint

## How to Generate Emotions in Animated Character:

- An actor playing in animation movie
- Face detected along with emotion/movement
- Create animated/synthetic character
- Imprint the emotion form original character

# Question: How Do you Detect A Face

- How do you detect a “face” if I give you the following image?

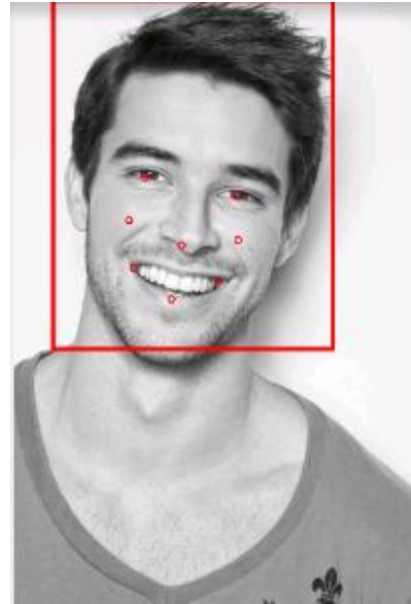


Image credit: <http://littledrun-k.tumblr.com/post/29367045597/>

# Challenges Associated with Face Detection

- **Pose (Out-of-Plane Rotation):**
  - looking away
- **Presence or absence of structural components:**
  - beard, mustache and glasses
- **Facial expression:**
  - face appearance is directly affected by a person's facial expression
- **Occlusion:**
  - faces may be partially occluded by other objects
- **Orientation (In-Plane Rotation):**
  - face appearance directly vary for different rotations about the camera's optical axis
- **Imaging conditions:**
  - lighting (spectra, source distribution and intensity) and camera characteristics (sensor response, gain control, lenses), resolution



# Face Detection Research Issues

- **Representation:** How to describe a typical face?
- **Scale:** How to deal with face of different size?
- **Search strategy:** How to spot these faces?
- **Speed:** How to speed up the process?
- **Precision:** How to locate the faces precisely?
- **Post processing:** How to combine detection results?

# Face Detector: Ingredients

- **Target application domain:** single image, video
- **Representation:** holistic, feature, etc.
- **Pre processing:** histogram equalization, etc.
- **Cues:** color, motion, depth, voice, etc.
- **Search strategy:** exhaustive, greedy, focus of attention, etc.
- **Classifier design:** ensemble, cascade
- **Post processing:** combining detection results

# Detecting Faces in Grayscale Images

- **Knowledge-based methods**
  - Encode human knowledge of what constitutes a typical face (usually, the relationship between facial features)
- **Feature invariant approaches**
  - Aim to find structural features of a face that exist even when the pose, viewpoint, or lighting conditions vary
- **Template matching methods**
  - Several standard patterns stored to describe the face as a whole or the facial features separately
- **Appearance-based methods**
  - The models (or templates) are learned from a set of training images which capture the representative variability of facial appearance

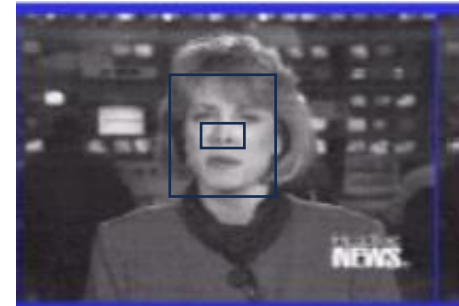
# Knowledge-Based Methods

- Top-down approach: Represent a face using a set of human-coded rules
- Example:
  - The center part of face has uniform intensity values
  - The difference between the average intensity values of the center part and the upper part is significant
  - A face often appears with two eyes that are symmetric to each other, a nose and a mouth
- Use these rules to guide the search process

# Knowledge-Based Methods

## Knowledge-based method [Yang and Huang 94]

- Multi-resolution focus-of-attention approach
- Level 1: (lowest resolution): apply the rule “the center part of the face has 4 cells with a basically uniform intensity” to search for candidates
- Level 2: local histogram equalization followed by edge detection
- Level 3: search for eye and mouth features for validation



# Knowledge-Based Methods

Knowledge-Based Method: [Kotropoulos & Pitas 94]

- Horizontal/vertical projection to search for candidates

$$HI(x) = \sum_{y=1}^n I(x, y) \quad VI(y) = \sum_{x=1}^m I(x, y)$$

- Search eyebrow/eyes, nostrils/nose for validation
- Difficult to detect multiple people or in complex background

# Knowledge-Based Methods: Summary

- **Pros:**

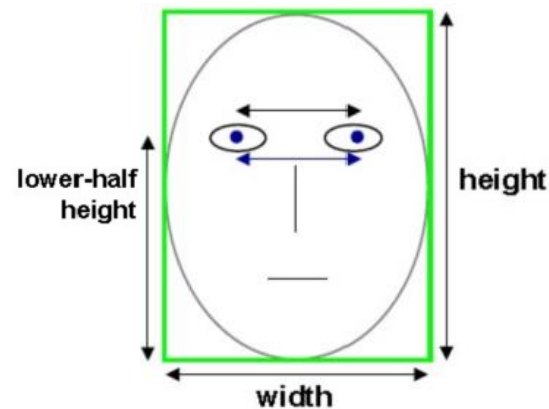
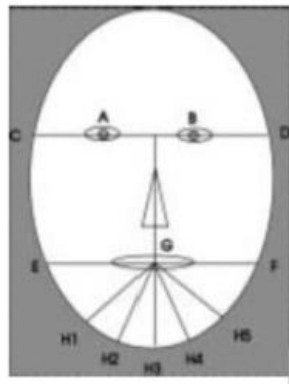
- Easy to come up with simple rules to describe the features of a face and their relationships
- Work well for face localization in uncluttered background

- **Cons:**

- Difficult to translate human knowledge into rules precisely: detailed rules fail to detect faces and general rules may find many false positives
- Difficult to extend this approach to detect faces in different poses: implausible to enumerate all the possible cases

# Feature-Based Methods

- Bottom-up approach: Detect facial features (eyes, nose, mouth, etc.) first
- Facial features: edge, intensity, shape, texture, color, etc.
- Aim to detect invariant features
- Group features into candidates and verify them



# Feature-Based Methods: Summary

- **Pros:**

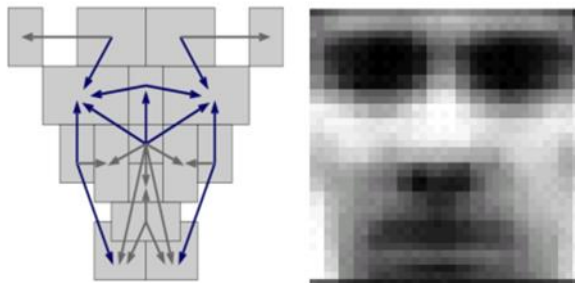
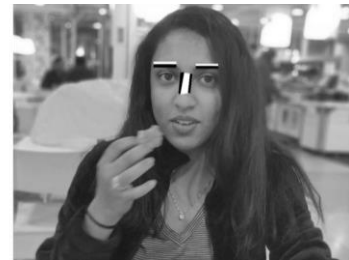
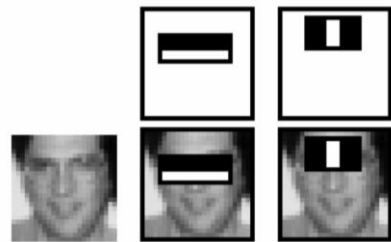
- Features are invariant to pose and orientation change

- **Cons:**

- Difficult to locate facial features due to several corruption (illumination, noise, occlusion)
- Difficult to detect features in complex background

# Template Matching Methods

- Template preparation
- Window by window matching between image segments and template
- One of the most commonly used methods



Ration Template [Sinha 94]

Template with ration or intensity



[Lanitis et al. 95]



average shape

Template with shape

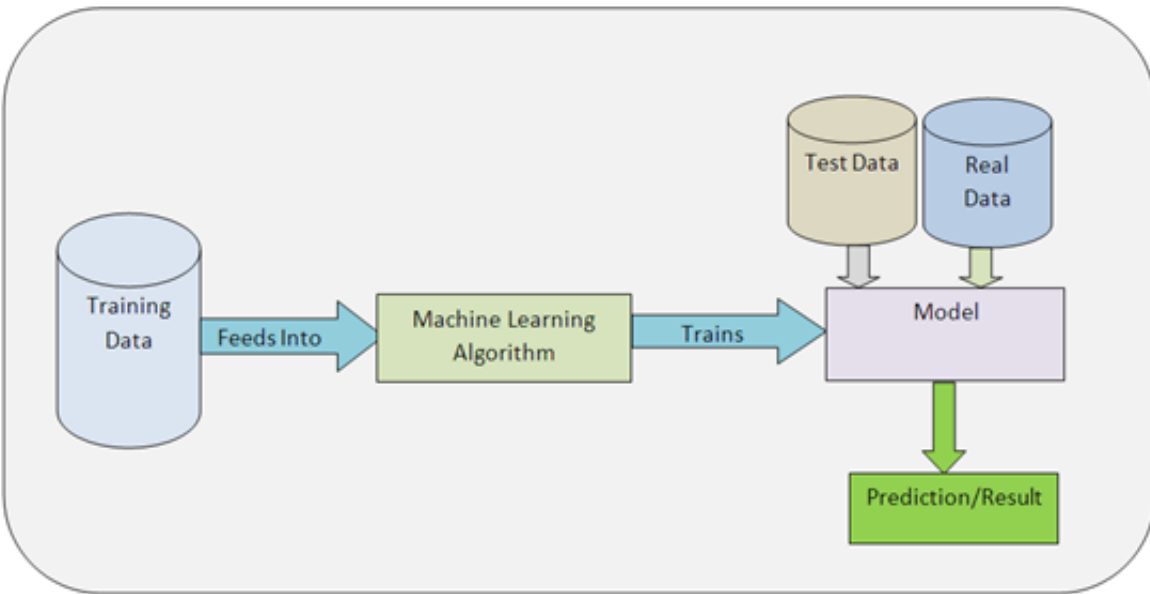
# Template Matching Methods: Summary

- Pros:
  - Simple
- Cons:
  - Templates needs to be initialized near the face images
  - Difficult to enumerate templates for different poses (similar to knowledge-based methods)

# Appearance Based Methods

- The models (or templates) are learned from a set of training images which capture the representative variability of facial appearance
- Methods differ in:
  - Representation
  - Preprocessing
  - Classifier Training
  - Search strategy
  - Post processing

# Appearance Based Methods: Generic Example



**A Simple Machine Learning Pipeline Explanation**

Image: <https://www.sqlservercentral.com/articles/understanding-machine-learning>

# Appearance Based Methods

- **Pros:**

- Use powerful machine learning algorithms
- Has demonstrated good empirical results
- Fast and fairly robust
- Extended to detect faces in different pose and orientation

- **Cons:**

- Usually needs to search over space and scale
- Need lots of positive and negative examples
- Limited view-based approach

# Detecting Faces in Color Images

**Idea:**

Use skin color information to separate faces from  
image background

# Detecting Faces in Color Images

## Methods of Skin Detection

- **Pixel-Based Methods**

- Classify each pixel as skin or non-skin individually, independently from its neighbors
- Simpler, faster
- Color Based Methods fall in this category

- **Region Based Methods**

- Try to take the spatial arrangement of skin pixels into account during the detection stage to enhance the methods performance
- More computation extensive
- Additional knowledge in terms of texture etc are required

# Detecting Faces in Color Images

## Skin Color based methods - Advantages

- Allows fast processing
- Robust to geometric variations of the skin patterns
- Robust under partial occlusion
- Robust to resolution changes
- Experience suggests that human skin has a characteristic color, which is easily recognized by humans

# Detecting Faces in Color Images

## Issues with skin color

- Are Skin and Non-skin colors separable?
- Illumination changes over time
- Skin tones vary dramatically within and across individuals
- Different cameras have different output for the identical image
- Movement of objects cause blurring of colors
- Ambient light, shadows change the apparent color of the image
- What color space to be used?

# Detecting Faces in Color Images

## Color-Based Face Detector: Summary

- **Pros:**
  - Easy to implement
  - Effective and efficient in constrained environment
  - Insensitive to pose, expression, rotation variation
- **Cons:**
  - Sensitive to environment and lighting change
  - Noisy detection results

# Detecting Faces in Color Images

## Video-Based Detectors: Summary

### **Pros:**

- An easier problem than detection in still images
- Use all available cues: motion, depth, voice, etc. to reduce search space

### **Cons:**

- Need for efficient and effective methods to process the multimodal cues
- Data fusion

# Conclusion

Face detection is not a solved problem.

Current focus is on detecting faces robustly under

- Varying poses
- Orientation
- Occlusion
- Expression
- Varying illumination
- Low image resolution

**Thank You!**

# Face Detection\_ Viola Jones

# Challenges with Face Detection

**What if we chose face templates/windows to search for faces in an image?**

- Sliding window detector must evaluate tens of thousands of location/scale combinations
- Faces are rare: 0–10 per image
  - For computational efficiency, we should try to spend as little time as possible on the non-face windows
  - A megapixel image has  $\sim 10^6$  pixels and a comparable number of candidate face locations
  - To avoid having a false positive in every image, our false positive rate has to be less than  $10^{-6}$

# Viola Jones

## The Viola/Jones Face Detector

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- A seminal approach to real-time object detection
- Training is slow, but detection is very fast

### Key Idea:

- Feature Evaluation with rectangular features
- Boosting for feature selection
- Cascades for fast rejection of non-face windows

P. Viola and M. Jones. [Rapid object detection using a boosted cascade of simple features](#). CVPR 2001.

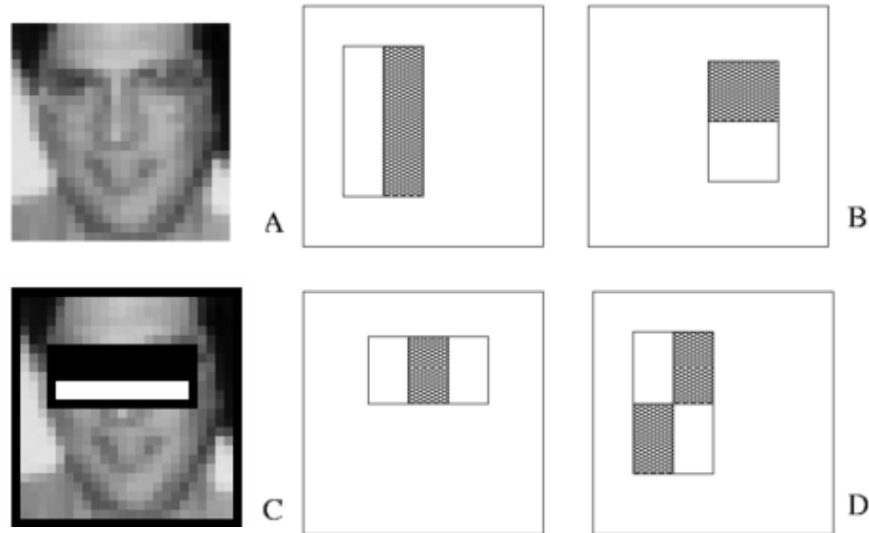
P. Viola and M. Jones. [Robust real-time face detection](#). IJCV 57(2), 2004.

# 1. Features/Kernel/Template Prep

## Key Idea:

- Rectangular features are used as “face features”

“Rectangle filters”

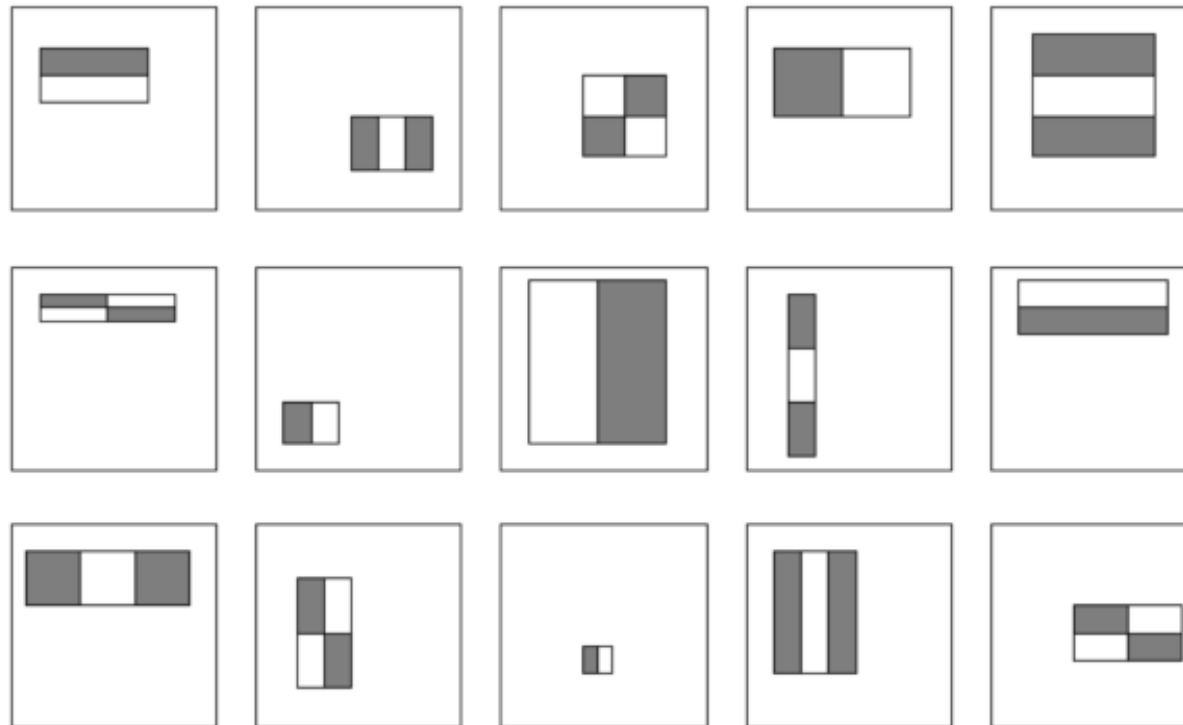


*Value =*

$$\sum (\text{pixels in white area}) - \sum (\text{pixels in black area})$$

# Feature Selection

- For a 24x24 detection region, the number of possible rectangle features is ~160,000!



# Feature Selection

- For a 24x24 detection region, the number of possible rectangle features is ~160,000!
- At test time, it is impractical to evaluate the entire feature set
- Can we create a good classifier using just a small subset of all possible features?
- How to select such a subset?

## 2. Intermediate Classification and Boosting

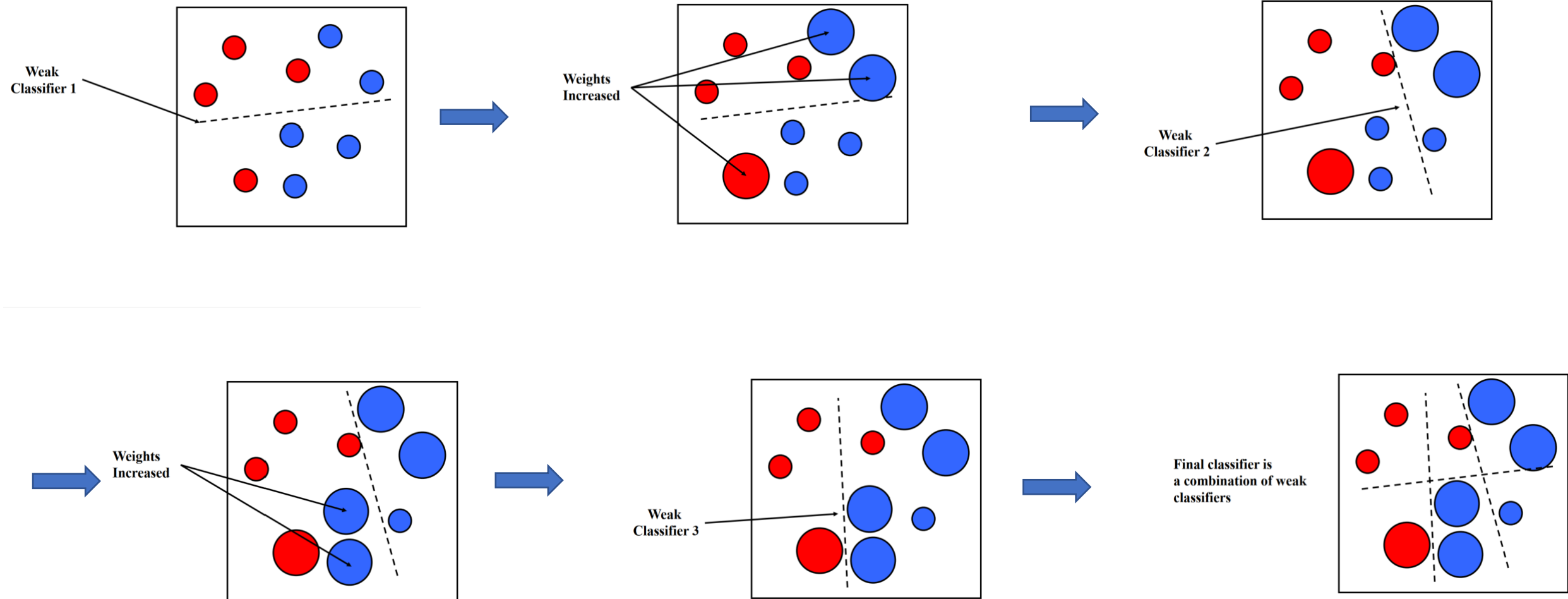
- Boosting is a classification scheme that works by combining *weak learners* into a more accurate ensemble classifier
- Training consists of multiple *boosting rounds*
  - During each boosting round, we select a weak learner that does well on examples that were hard for the previous weak learners
  - “Hardness” is captured by weights attached to training examples

# Training Procedure

- Initially, weight each training example equally
- In each boosting round:
  - Find the weak learner that achieves the lowest *weighted* training error
  - Raise the weights of training examples misclassified by current weak learner
- Compute final classifier as linear combination of all weak learners (weight of each learner is directly proportional to its accuracy)
- Exact formulas for re-weighting and combining weak learners depend on the particular boosting scheme (e.g., AdaBoost)

Y. Freund and R. Schapire, [A short introduction to boosting](#), *Journal of Japanese Society for Artificial Intelligence*, 14(5):771-780, September, 1999.

# Boosting Illustration

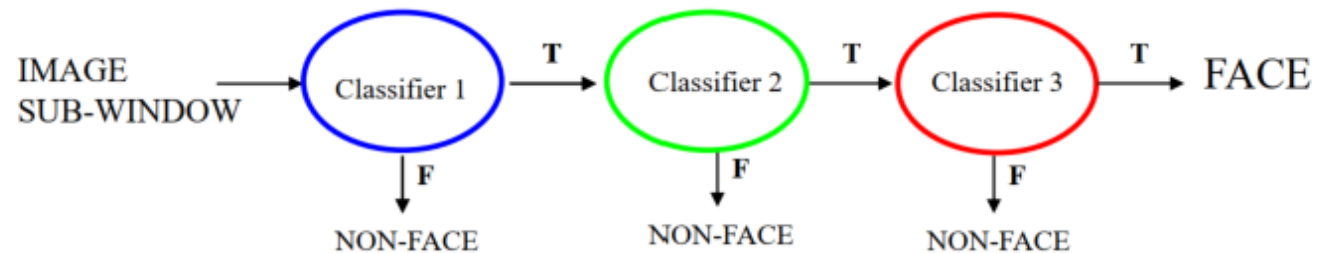


# Summary

- Advantages of boosting
  - Integrates classification with feature selection
  - Flexibility in the choice of weak learners, boosting scheme
  - Testing is fast
  - Easy to implement
- Disadvantages
  - Needs many training examples

# Cascade

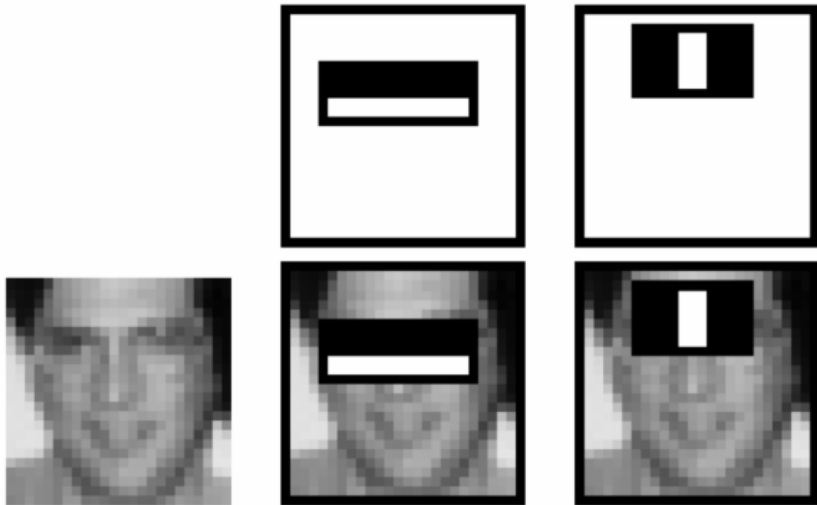
- We start with simple classifiers which reject many of the negative sub-windows while detecting almost all positive sub-windows
- Positive response from the first classifier triggers the evaluation of a second (more complex) classifier, and so on
- A negative outcome at any point leads to the immediate rejection of the sub-window



- Viola Jones- originally meant for object detection
  - Try using it for 'any' object detection
  - Let's see the results on face detection

# Boosting for face detection

- First two features selected by boosting:



This feature combination can yield 100% detection rate and 50% false positive rate



# Experiment

- Training Data
  - 5000 faces
    - All frontal, rescaled to 24x24 pixels
  - 300 million non-faces
    - 9500 non-face images
  - Faces are normalized
    - Scale, translation
- Many variations
  - Across individuals
  - Illumination
  - Pose

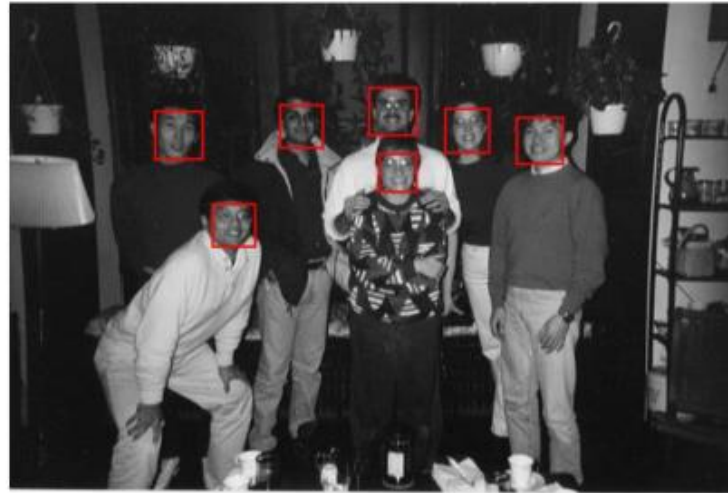
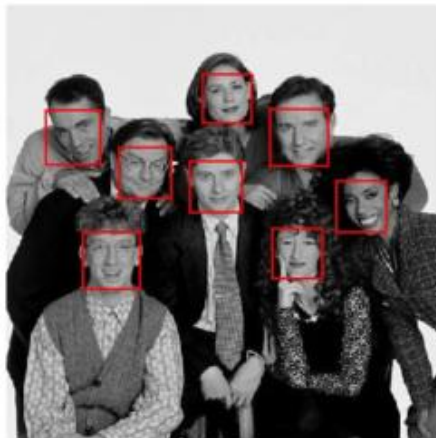


# System Performance

- Training time: “weeks” on 466 MHz Sun workstation
- 38 layers, total of 6061 features
- Average of 10 features evaluated per window on test set
- “On a 700 Mhz Pentium III processor, the face detector can process a 384 by 288 pixel image in about .067 seconds”
  - 15 Hz
  - 15 times faster than previous detector of comparable accuracy (Rowley et al., 1998)

# Output

## Output of Face Detector on Test Images



# Conclusion

## Summary: Viola/Jones detector

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- Rectangle features
- Boosting for feature selection
- Attentional cascade for fast rejection of negative windows

**Thank You!**